

DPCN Assignment 1: Opinion Network Formation from an Ordinal Attitude Questionnaire

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Abstract

We construct and analyse an opinion network from a 60-item ordinal attitude questionnaire answered by 96 respondents across four thematic blocks (technology, education, society, environment). The work has three parts. First, we sanitise the data while preserving the distinction between two different missingness mechanisms: structural blanks left when a respondent abandons the survey, and scattered “No Comments” (NC) responses inside otherwise complete questionnaires. NC is not recoded as Neutral and NC users are not discarded; missing responses are instead recovered by question-wise regularised proportional-odds ordinal logistic regression, validated by artificial masking under both missingness patterns. Second, we show that the response distribution is severely ceilinged—79.8% of valid answers fall on the agree side and the grand mean is +1.14 on a $[-2, +2]$ scale—so that raw similarity measures how agreeable a respondent is rather than what they believe. Row-centring each respondent removes this component and reduces the number of item pairs correlated at $|r| \geq 0.30$ from 327 to 42. Third, we build correlation networks over the sanitised matrix and run Deffuant–Weisbuch bounded-confidence dynamics on a respondent network. We show analytically and by controlled experiment that the widely reported cluster plateau at small confidence thresholds is an arithmetic artefact of the five-point response grid, while the difference between the sparse network and a well-mixed population is a genuine effect of topology.

1 Introduction

Likert-scale questionnaires are a natural setting for studying how attitudes in different domains relate to one another. Although each item is answered independently, responses exhibit substantial dependence across questions and across thematic domains. Network representations model these dependencies directly: questionnaire items (or respondents) become vertices, and statistical association becomes weighted edges.

Turning a questionnaire into a network is not, however, a mechanical step. Three decisions determine whether the resulting graph carries signal or merely re-describes the response format.

How missing data is handled. Missing responses arise both from respondents stopping before reaching later questions and from respondents declining to answer a particular item. These are different mechanisms [10] and should not be silently merged, and assigning a numerical value such as Neutral to a declined answer injects information the respondent never supplied.

Whether response style is removed. When respondents agree with almost everything, correlation between any two items is inflated by a shared “agreeableness” component rather than genuine item association [9]. Unless this is removed, the network largely encodes response style.

What the network is compared against. A correlation-derived network has transitivity built into its construction, so clustering and modularity appear even in structureless data. Any structural claim needs a null model beside it.

This report addresses all three. Section 2 documents the dataset and its ceiling effect, Section 3 the sanitisation, imputation and row-centring pipeline, Section 4 the individual analyses and visualisations, and Sections 5–6 the findings and their limits.

Team and code. Team [TEAM NAME]. All code, figures, and the data pipeline are available at https://github.com/shreyash-lohare/dpcn_a1_opinion_network. The full analysis is reproducible from a fresh clone via `python run_all.py` followed by the figure modules in `src/`.

2 Dataset Documentation

The questionnaire contains 60 items in four thematic blocks of 15: technology (T01–T15), education (E01–E15), society and ethics (S01–S15), and environment (V01–V15). Each item offers five ordered categories plus an explicit No Comments option. For numerical analysis the categories are encoded symmetrically,

$SD = -2, \quad D = -1, \quad N = 0, \quad A = +1, \quad SA = +2,$

which represents the ordering of the categories without asserting that adjacent categories are equally spaced. Descriptive statistics in this section are reported on the original 1–5 scale (subtract 3 to recover the encoding above).

The raw file contains 96 respondent records and $96 \times 60 = 5,760$ cells, of which 5,255 were answered and 5,218 carry a substantive ordinal response. Blocks are internally consistent to varying degrees: Cronbach’s α [3] is 0.87 for environment and 0.82 for society, but only 0.65 for technology and 0.58 for education, and 0.885 across all 60 items.

2.1 The response ceiling

The single most consequential property of this dataset is its ceiling. Figure 1(a) shows that 79.8% of valid answers are Agree or Strongly Agree, with Strongly Agree alone accounting for 43.1%; only 7.0% of answers fall on the disagree side. The grand mean is 4.138 on the 1–5 scale, i.e. +1.14 on the $[-2, +2]$ encoding.

Two consequences follow, and both shape every later decision. Item means barely discriminate between statements, so the informative signal is in which statements attract disagreement rather than in overall agreement level. And because almost every respondent agrees with almost everything, two respondents will look similar under a raw similarity measure simply by both being agreeable. Section 3.3 removes this component explicitly.

Figure 1(b) shows the ceiling is not uniform across topics. Environment ($\mu = 4.39$, 89.5% agree) and society ($\mu = 4.28$, 86.0%) are close to settled consensus, whereas technology ($\mu = 3.99$, 8.1% disagree) and especially education ($\mu = 3.91$, 13.8% disagree) retain real disagreement. Notably, the blocks with the most disagreement are also those with the lowest internal consistency, consistent with their items tapping several distinct opinions rather than one settled disposition.

2.2 Where the disagreement lives

Figure 2 ranks the twelve items attracting the most disagreement. Seven are education items and four are technology items; only one comes from society, and none from environment. Compulsory class attendance (E03, mean 2.06) is the only statement the cohort actively rejects, with 70.1% disagreeing and 40% strongly disagreeing. Written examinations as a measure of knowledge (E02, mean 2.84) is the next most rejected.

The pattern is specific rather than generally oppositional: the same cohort returns 94.3% agreement that curricula should be updated more often (E11) and that universities should prioritise problem-solving over rote

learning (E10). What is contested is assessment and teaching format, not the institution.

The technology items behave similarly. Teaching AI (T04, 81.3%), AI as a learning aid (T02, 79.1%), and AI accelerating research (T05, 81.3%) all score high, whereas AI-assisted medical diagnosis (T08, mean 3.08) and autonomous vehicles (T09, mean 3.28) are the two lowest technology items by a clear margin. Since robots in hazardous occupations (T07) scores 82.4% agreement, the objection is not to automation in general but to delegating consequential judgment.

2.3 Missingness and data-quality notes

Of the 96 records, 85 are complete, 5 are entirely empty, 4 stop after the technology block, and 2 stop partway through environment. Dropout is therefore monotone—a truncated tail rather than scattered gaps—which matters because it means later blocks rest on a slightly smaller base ($n = 84$ –91) than earlier ones. Separately, 37 cells carry No Comments, scattered across 17 respondents who otherwise completed the questionnaire.

Three limitations should be carried into every result below. Items with means above roughly 4.5 measure social-desirability agreement more than strength of preference. Only a small minority of statements are worded to invite disagreement from an otherwise positive respondent, so acquiescence bias and a genuine ceiling cannot be fully separated here. And the file contains no demographic fields, so no finding can be attributed to a subgroup.

3 Data Handling

3.1 Dataset sanitation

Missingness structure. Five entirely empty records are excluded, leaving 91 respondents. No respondent is excluded for being merely incomplete: the four who stopped after technology and the two who stopped during environment are retained, and their unanswered items are treated as missing data to be recovered rather than as grounds for deletion. This yields 205 contiguous structural blank cells and 37 NC cells, 242 missing cells in total among the retained respondents.

We distinguish

structural blank = question not reached,

from

NC = deliberate non-response.

Both become missing values during modelling, but their original types are retained so that preprocessing never silently conflates abandonment with an explicit decision to withhold an answer.

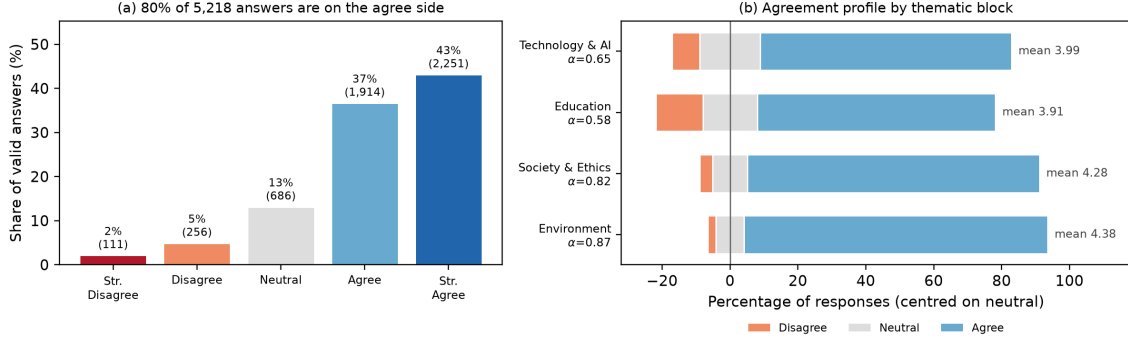


Figure 1: The response ceiling. (a) Distribution of all 5,218 valid answers: nearly four in five are on the agree side and 43% are Strongly Agree. (b) Agreement profile per block, centred on neutral, with block mean and Cronbach’s α . Environment and society are near consensus; education carries the most disagreement and the weakest internal consistency.

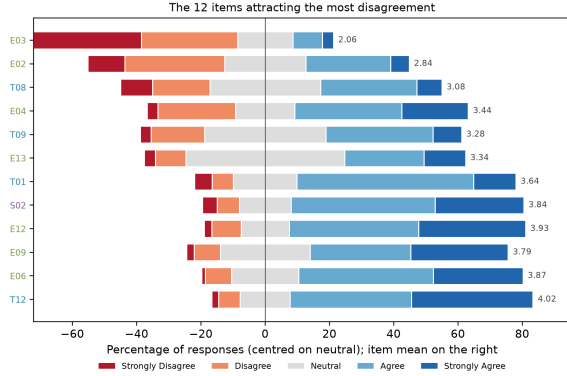


Figure 2: The twelve items attracting the most disagreement, as diverging response profiles centred on neutral, with item mean on the right. Labels are coloured by block. Disagreement concentrates in education (7 of 12) and technology (4 of 12).

Why NC is not Neutral. Recoding NC as Neutral would assign a substantive response to a cell for which none was supplied. As a diagnostic we fitted a two-way additive model with respondent and item effects on the clean responses, $\hat{x}_{iq} = \bar{\mu} + a_i + b_q$. The fitted value at the 37 NC cells averaged 3.92 on the 1–5 scale against 3 for Neutral, so recoding would impose a systematic downward shift of about 0.92 scale points. This model is used only as a diagnostic, not as the imputation method.

Empirical NC diagnostics. NC usage shows no systematic item- or respondent-level structure under the tests applied. NC counts per item have a dispersion index of 1.10 and a chi-squared test of item-level uniformity gives $p = 0.27$; no item carries more than three NC responses; the correlation between NC frequency

Table 1: Sensitivity of the item-correlation network to NC treatment at $\tau = 0.30$, against pairwise deletion as reference. Computed at an earlier 86-respondent configuration; ordering of the treatments is the result of interest rather than the absolute counts.

Treatment	N	E	r	Jacc.	AMI
Pairwise delete (ref.)	86	325	—	—	—
Recode $\rightarrow$ Neutral	86	333	.993	.875	.662
Drop NC users	69	362	.947	.616	.337
Two-way impute	86	335	.997	.924	.670

and item standard deviation is $r = -0.02$. At respondent level, NC users have mean score 4.01 against 4.15 for non-users ($p = 0.27$), and neutral-response rates of 0.123 against 0.142 ($p = 0.82$). There is therefore no empirical basis for assigning NC a fixed ordinal category, and we retain it as unobserved.

Sensitivity to NC treatment. Table 1 compares four NC treatments on the item-correlation network at $\tau = 0.30$. Dropping NC users would remove 17 respondents, roughly a fifth of the sample, to eliminate only 37 cells, and it is the only treatment that materially disturbs the recovered community structure (AMI 0.337). Correlations are far more stable than partitions across all four treatments—a warning that stability of a correlation matrix should never be read as evidence that a particular partition of it is stable.

3.2 Missing-value imputation

Ordinal regression formulation. For target item j the remaining 59 items form the predictor vector $\mathbf{x}_{i,-j}$, and the target satisfies $y_{ij} \in \{-2, -1, 0, +1, +2\}$. We fit one regularised proportional-odds ordinal logistic

Table 2: Artificial-masking validation. Accuracy is exact-category; MAE is mean absolute error on the five-point scale.

Masking	Ordinal model		Mode baseline	
	Acc.	MAE	Acc.	MAE
Scattered (NC-like)	.539	.534	.486	.626
Suffix (structural)	.498	.612	.482	.655

regression [7] per item, with latent score $z_i = \beta_0 + \beta^\top \mathbf{x}_{i,-j}$ and ordered thresholds $\theta_1 < \theta_2 < \theta_3 < \theta_4$ inducing the five response intervals. The prediction is the maximum-probability category,

$$\hat{y}_{ij} = \arg \max_k P(y_{ij} = k \mid \mathbf{x}_{i,-j}).$$

This preserves the ordering of the Likert responses while avoiding the interval-scale assumption of linear regression, and unlike nominal multiclass classification it retains the relative order of the five outcomes.

Missing predictors and regularisation. A respondent may have several missing cells, so predictors that are themselves missing cannot be encoded as Neutral. They are instead filled with the training-fold column median before standardisation, with the statistic computed only from training data in each fold to prevent leakage from held-out responses. A ridge penalty controls complexity in the 59-dimensional predictor space, and the target is always excluded from its own predictor vector, giving $M_j : \mathbf{X}_{-j} \rightarrow Y_j$ for $j = 1, \dots, 60$.

Validation. The true values of the 242 genuinely missing cells are unavailable, so the pipeline is evaluated by artificial masking of responses whose values are known, under two mechanisms matched to the two missingness types. Scattered masking hides 37 observed cells at random inside otherwise complete respondents, approximating NC, over 40 trials. Suffix masking hides a contiguous tail after a simulated stopping point drawn from the observed stopping pattern, for 15 respondents per trial over 15 trials; items after the stopping point are withheld from the predictor so the model cannot use information unavailable in practice.

Because the target is ordinal, MAE is the more informative metric: a prediction one category away is not the same error as one at the opposite end of the scale. Table 2 shows the model beats a modal-response baseline under both mechanisms, by a wider margin under scattered masking where more co-answered items remain available. Performance is lower under suffix masking precisely because fewer predictors survive the stopping point, so the two settings should be read separately rather than averaged into one number. The

final model imputes all 242 missing cells, with every recovered value restricted to the original five-level support.

3.3 Row-centring

Imputation yields a complete 91×60 matrix on the original scale, but not one that can be correlated directly. Because the grand mean is +1.14 (Section 2), a respondent who agrees with everything raises all 60 of their entries together, which appears as association between every pair of items at once. We therefore replace each entry by its deviation from that respondent’s own mean,

$$x'_{i,j} = x_{i,j} - \bar{x}_{i\cdot}, \quad \bar{x}_{i\cdot} = \frac{1}{60} \sum_{k=1}^{60} x_{i,k}. \quad (1)$$

Two points are worth stating precisely. First, centring is applied strictly after imputation and strictly before correlation: the imputation model predicts on the original -2 to $+2$ scale, and centring is a network-construction step, not a data-cleaning one. Second, rank correlation does not remove this bias automatically—Spearman correlation is invariant to a monotone transform of a single column, not to an additive shift that differs from row to row. The magnitude of the effect is reported as Graph 3 (Section 4.3).

3.4 Network construction

On the sanitised, row-centred matrix we build a weighted undirected graph $G = (V, E, W)$ whose 60 vertices are the questionnaire items. Because responses are ordered categories, the primary association measure is the Spearman rank correlation ρ_{qr} , and edge weights are signed correlations, with $|\rho_{qr}|$ used for edge magnitude in visualisation and sign distinguished by colour. We retain the full weighted matrix and threshold only for display, showing an edge when $|\rho_{qr}| \geq \tau$. The association step is implemented modularly so a latent-variable ordinal measure such as polychoric correlation can replace Spearman without changing graph construction. A second, respondent-level network is introduced in Section 4.12 for the dynamics experiment.

4 Analysis and Visualisations

This section reports the twelve analyses. Each states what the figure shows, what was found, and what follows from it. Sections still owned by other team members are marked as placeholders.

A note on respondent counts. Several reference values circulated during planning assume a pipeline variant that drops respondents with more than 12 missing items, leaving 86. This report does not use that filter: all 91 non-empty respondents are retained and their missing cells recovered by ordinal regression (Section 3.2). Figures produced here therefore report $n = 91$, and small deviations from those reference numbers are expected.

4.1 Graph 1: Missingness map and response distribution

Figure 3 shows every cell of the raw 96×60 matrix as missing or present, with respondents sorted by missing count, beside the distribution of per-respondent mean scores.

Findings. The 542 missing cells (505 structural blanks plus 37 NC) are extremely unevenly distributed. Five respondents are entirely empty and account for 300 of them; the remaining 242 are spread thinly across the other 91, a missing rate of 4.4%. Below the cut-off the map is almost clean, with scattered single cells and two visible tails from the respondents who stopped early. The right panel locates the grand mean at 1.14, with the bulk of respondents between 0.75 and 1.50 and only one respondent below zero.

Implication. The missingness structure justifies the retention policy directly. Because all but five respondents answered at least 80% of items, there is no population of badly-incomplete respondents whose removal would simplify the analysis, and a count-based exclusion rule would discard usable respondents to remove very few cells. The right panel is the first direct evidence of the acquiescence problem that Section 3.3 corrects.

4.2 Graph 2: Item variance ranking

Figure 4 ranks all 60 items by standard deviation on the uncentred imputed matrix. Centring is deliberately not applied here: it changes each item’s variance and would answer a different question (spread after removing each respondent’s baseline) than the one asked (how divided the cohort is on the statement as posed).

Findings. Item SD ranges from 0.50 (E15, “continuous learning is essential”, near-unanimous) to 1.18 (E04, online learning). The five most contested items are E04, E03, E02, T08 and T07; the least contested include V13, V06, S05 and E15. By block mean SD the order is technology 0.92, education 0.88, society 0.79, environment 0.72.

Implication. The block averages and the extremes tell slightly different stories, and both matter. Technology has the highest average variance, but the sharpest disagreements are educational: three of the five most

divided items are E02–E04, all concerning assessment and teaching format. Environment contributes none of the top ten. This is a property of the survey instrument, not of preprocessing: the questionnaire contains a settled domain (environment), a moderately settled one (society), and two live ones (technology, education). Consensus items carry little information for a correlation network—an item everyone endorses cannot covary with anything—so the network’s signal is necessarily concentrated in the contested tail.

4.3 Graph 3: The effect of row-centring

Figure 5 overlays the distribution of all $\binom{60}{2} = 1,770$ pairwise item correlations computed before and after row-centring, on identical bin edges.

Findings. The raw distribution is displaced positive, with mean $r = 0.151$ and median 0.151; almost no pair is appreciably negative. After centring the distribution sits essentially on zero, mean $r = -0.004$, median -0.008 , with positive skew (0.221) leaving a heavier right tail of genuinely co-varying pairs. At a fixed threshold of $|r| \geq 0.30$ the number of surviving pairs falls from **327** to **42**, a reduction of 87%.

Implication. This converts the claim of Section 3.3 from an assertion into a measurement: roughly seven eighths of the apparent association in the raw matrix is shared response style, not item relationship. Had the network been built on raw scores it would have been dense, almost entirely positive, and would have supported confident but spurious conclusions about item structure. The surviving 42 pairs are the only ones worth interpreting, and the emergence of negative correlations after centring—impossible to see in the raw distribution—is what later permits a signed-network analysis at all.

4.4 Graph 4: Real versus shuffled correlations

Figure 6 compares the observed distribution of all 1,770 item–item correlations against a null that destroys inter-item association while preserving every item’s marginal distribution exactly. Each of 200 replicates permutes every column independently and then re-runs the full pipeline—row-centre, then correlate—on the shuffled matrix. This detail is not cosmetic: applying a single permutation to all columns, or permuting rows, merely relabels respondents and leaves the correlation structure numerically untouched.

Graphs 4, 5 and 7 are computed on the frozen 86-respondent matrix rather than the 91-respondent ordinal-imputed matrix of Section 3.2. Section 5.3 reports what changes under the other choice.

Findings. Correlations estimated from $n = 86$ respondents scatter about zero with standard deviation $1/\sqrt{n-3} = 0.110$ even when nothing is related. The

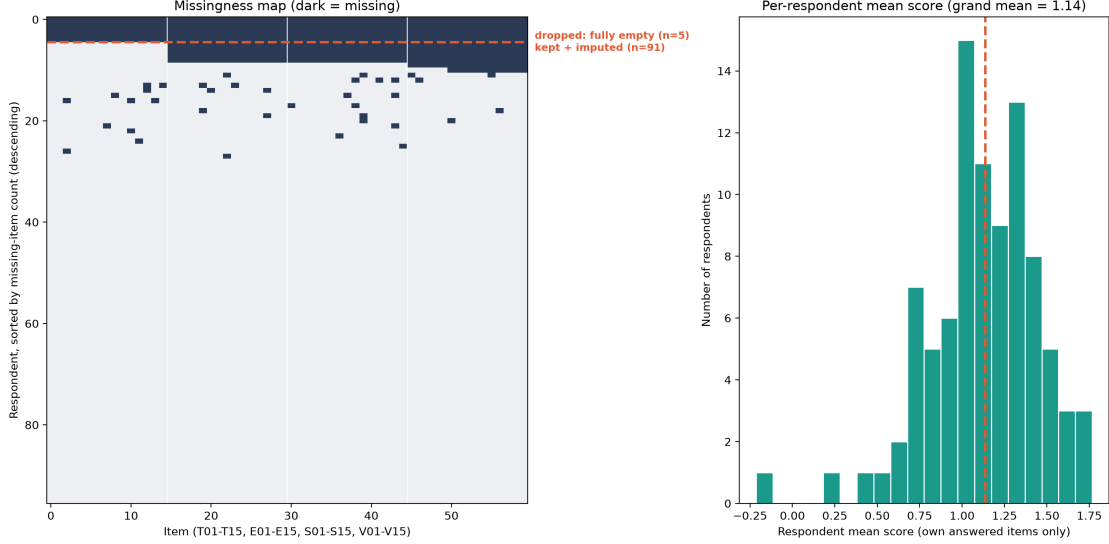


Figure 3: Graph 1. Left: missingness map over all 96 raw respondents, sorted by missing-item count; dark cells are missing for either reason. The dashed line marks this project’s exclusion rule—only the five entirely empty records are dropped, and the remaining 91 respondents have their 242 missing cells imputed. Right: per-respondent mean over answered items only, with the grand mean at 1.14.

permuted null reproduces this almost exactly at **0.108**, against **0.136** for the observed matrix. Real structure exists, but it is modest, and it lives in the tails rather than the bulk:

$ r >$	0.15	0.20	0.25	0.30
Observed edges	467	258	111	49
Expected by chance	308	119	37	9
Chance fraction	66%	46%	33%	19%

Implication. At the conventional cutoff $|r| > 0.15$ roughly two thirds of surviving edges are what 1,770 simultaneous estimates from 86 people produce by luck alone. A network drawn there is mostly noise with a real signal embedded in it, and no amount of downstream community detection can separate the two after the fact. This figure is therefore a precondition for Section 4.7: it converts threshold selection from a matter of convention into a measured trade-off, and it supplies the chance-expectation curve that the sweep is read against.

4.5 Graph 5: Eigenvalue spectrum against Marchenko–Pastur

Graph 4 establishes that a noise floor exists but not how many genuine dimensions sit above it. Principal component analysis alone cannot settle this: PC1 explains only 8.3% of variance, and that number is equally consistent with weak structure and with none. Marchenko–Pastur theory [5] answers it exactly. For a correlation matrix built from T observations of N

pure-noise variables, the eigenvalues fall inside an analytically known band, and anything above its upper edge carries more structure than T observations could have produced by chance.

Findings. With $T = 86$ and $N = 60$, $Q = T/N = 1.433$ and $\lambda_+ = (1 + 1/\sqrt{Q})^2 = \mathbf{3.368}$. Figure 7 shows the bulk of the spectrum tracking the theoretical density closely, with exactly three eigenvalues clearing the band—4.05, 3.98 and 3.53—together carrying **19.3%** of total variance. Reading the item loadings, the three modes correspond to (i) systemic responsibility against technological optimism and traditional instruction, (ii) regulatory guardrails against individual autonomy, and (iii) AI-enabled education and industry against environmental stewardship. The third is a clean split; the first two mix items across blocks and we present them as readings of the loadings rather than as established constructs.

One caveat belongs beside the headline. The fourth eigenvalue is 3.354 against a band edge of 3.368—a margin of 0.4%. “Exactly three” is the correct count, but it is not a comfortable one, and a slightly different respondent filter changes it (Section 5.3).

Implication. Four fifths of this correlation matrix is sampling noise. That is the single most important quantitative statement in the analysis, and it disciplines everything downstream: the network is rebuilt from the three significant modes alone, with the diagonal restored to unity, so that later community and balance analyses operate on the part of the matrix that 86 respondents can actually support. It also reinforces

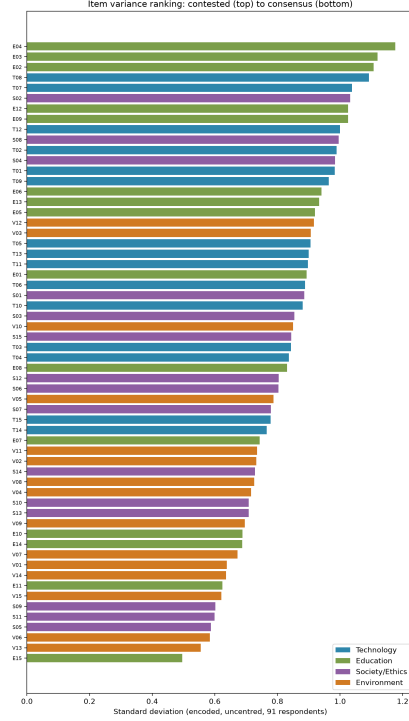


Figure 4: Graph 2. All 60 items ranked by standard deviation on the uncentred imputed matrix, coloured by block. Contested items (top) concentrate in education and technology; environment items cluster at the consensus end.

the reading of Section 5: three modest, cross-cutting dimensions carrying under a fifth of variance is the signature of a broadly agreeing cohort with local structure, not of a polarised one.

4.6 Graph 6: Reordered correlation heatmap

[TO BE COMPLETED — Graph 6 — owner: P3]
 Correlation matrix with rows and columns reordered by detected community so that communities appear as diagonal blocks, with a colour scale that distinguishes sign.
 Findings + implication: to be written. State explicitly whether the diagonal blocks align with the four thematic blocks or cut across them — that comparison is the point of the figure.

4.7 Graph 7: Threshold and percolation sweep

Reporting a network at one threshold invites the obvious objection that the threshold was chosen to produce the result. Figure 8 answers it by rebuilding the signed graph at all 46 cutoffs from 0.05 to 0.50 and tracking edge count, giant-component fraction, fragmentation and Louvain modularity [1, 8] across the whole range, so the chosen value is visibly a point on a curve. The

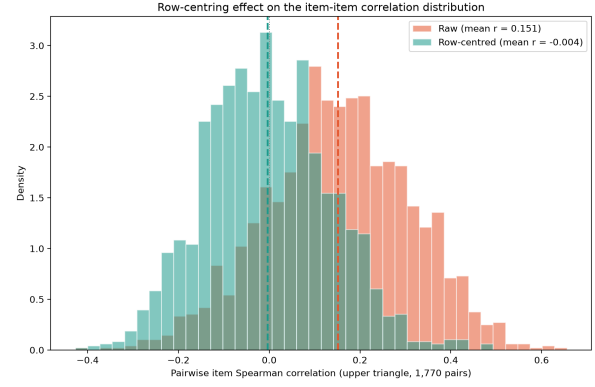


Figure 5: Graph 3. Distribution of all 1,770 pairwise item correlations before and after row-centring, on shared bin edges. Centring moves the mean from 0.151 to -0.004 and removes 87% of the pairs exceeding $|r| \geq 0.30$.

sweep also absorbs the giant-component analysis entirely.

Findings. The two constraints pull in opposite directions. Raising the threshold lowers the chance fraction from Graph 4, but the graph shatters well before the chance fraction becomes small: by $|r| > 0.25$ the giant component holds fewer than half the items and 32 of 60 are isolated. We therefore state the rule explicitly—the highest cutoff at which the giant component still spans more than two thirds of the items and mean degree remains above 4, so that community and path-based metrics stay meaningful—and note that spectral denoising has already removed the sampling noise the chance column measures, so raising the cutoff further would penalise the same noise twice. This selects $|r| > 0.20$, giving **124** edges (60 positive, 64 negative), mean degree 4.13, a giant component of 41 of 60 items (68%), and modularity $Q = 0.405$.

Two independent controls agree at this cutoff despite sharing no machinery. The permutation null implies the raw graph’s 258 edges contain about 119 chance edges, leaving roughly 139 real ones; spectral denoising, which never sees the null, retains 124. The estimates differ by about 11%.

Implication. Panel (d) is the cautionary one. Modularity climbs steadily to 0.63 as the threshold rises—but at that point the graph is 51 components with 47 isolated items. Quoting a single high modularity without the sweep beside it would report fragmentation as community structure, which is precisely the error the null models of Section 5.2 are designed to catch.

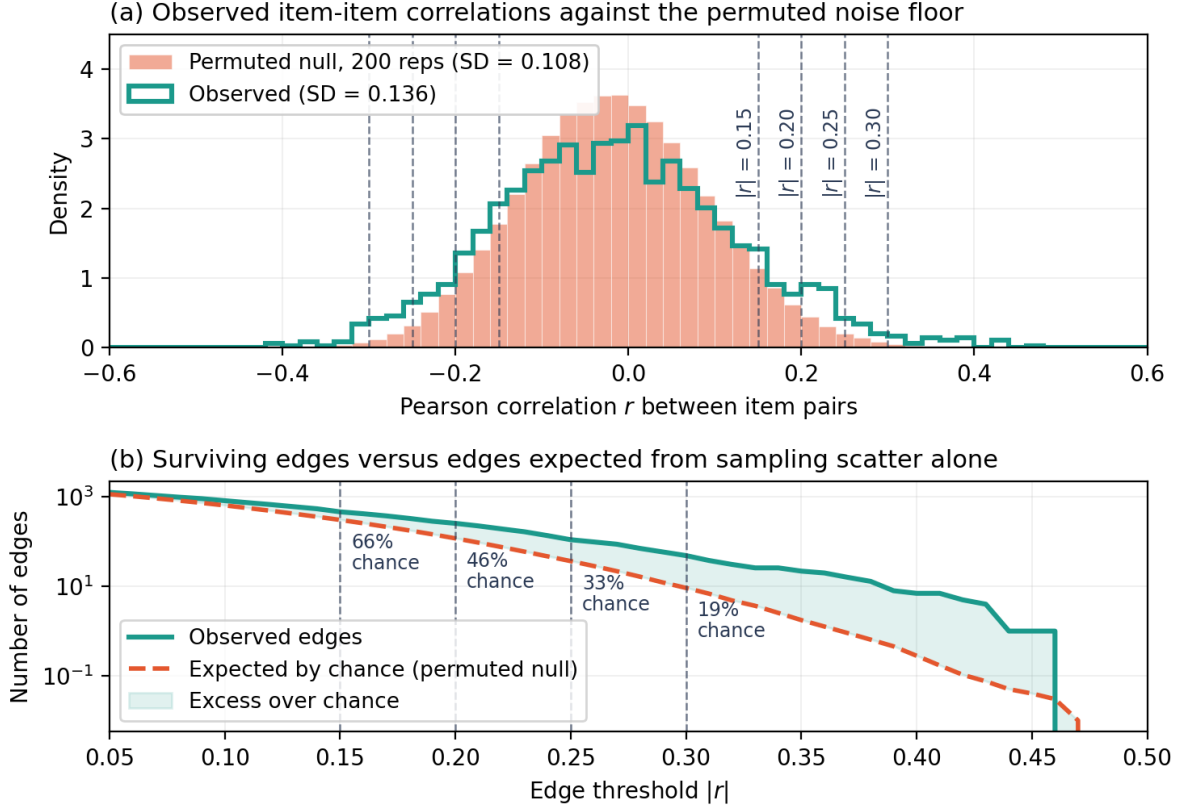


Figure 6: Graph 4. (a) Observed item-item correlations against a 200-replicate permutation null that re-runs the whole pipeline on independently shuffled columns; observed SD 0.136 against 0.108 for the null, with the excess confined to the tails. (b) Surviving edges against edges expected from sampling scatter alone, log scale; the shaded band is the excess over chance. At $|r| > 0.15$ two thirds of surviving edges are chance.

4.8 Graph 8: Main item network

[TO BE COMPLETED — Graph 8 — owner: P1, blocked on Graph 7]

The item-item network at the threshold selected in Graph 7, nodes coloured by thematic block, edges signed and weighted by correlation magnitude.

Dependency: the threshold must come from the Graph 7 sweep rather than being chosen here.

Findings + implication: to be written. Report the number of connected items and isolates at the chosen threshold, which items act as hubs, and whether edges respect block boundaries.

4.9 Graph 9: Topic-block connectivity

The questionnaire arrives with its own partition: fifteen items each for technology, education, society and environment. That partition is an editorial claim about what belongs together. Whether it is also a statistical one is a separate question, and it is the first thing the network should be asked, because every later community result is interpreted against it. We collapse the

60×60 item correlation matrix into a 4×4 block matrix by averaging within each block pair. Off-diagonal cells average all 225 cross-block pairs; diagonal cells average only the 105 distinct within-block pairs, with self-correlations excluded—retaining the unit diagonal would add a constant to every diagonal cell and manufacture exactly the block structure being tested for.

Significance comes from a block-label permutation: the correlation matrix is held fixed and the T/E/S/V labels are reshuffled across the 60 items, preserving the 15/15/15/15 design, over 5,000 seeded draws. This is deliberately not the response-permutation null of Section 4.4, which regenerates the whole pipeline to ask whether the network is meaningful at all. The question here is narrower and needs no threshold: given this matrix, is the survey's own partition better aligned with it than an arbitrary partition of the same shape? Because only labels move, the test is immune to any constant offset in the matrix.

Findings. Row-centring is what makes the block structure visible at all. On the raw matrix (Figure 9, left) every one of the sixteen cells is positive, from

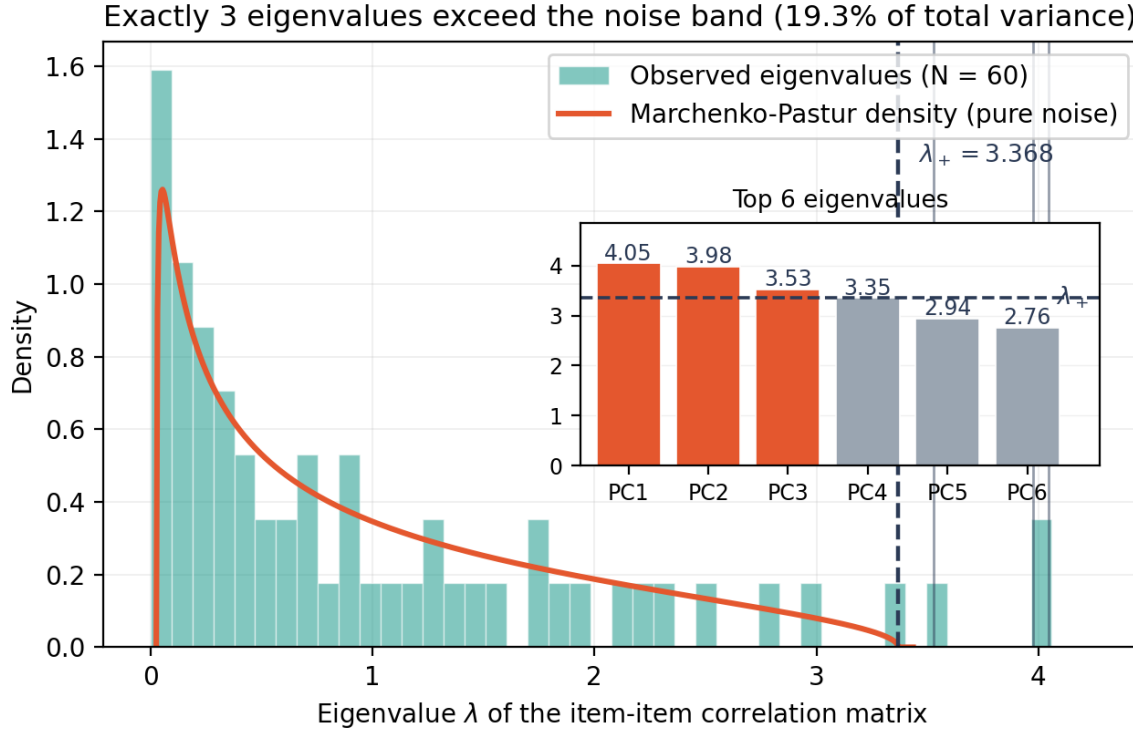


Figure 7: Graph 5. Eigenvalue spectrum of the row-centred item correlation matrix against the Marchenko–Pastur density for pure noise of the same shape ($T = 86$, $N = 60$, $Q = 1.433$). The bulk tracks the theoretical curve; exactly three eigenvalues exceed $\lambda_+ = 3.368$, carrying 19.3% of total variance. Inset: the top six eigenvalues, showing the fourth (3.354) falling just below the band edge.

+0.041 to +0.377: acquiescence makes all four topics appear mutually connected, and the thematic design is invisible underneath it. On the row-centred matrix (centre) the sign pattern separates cleanly—every diagonal cell positive, every off-diagonal cell negative. The partition is strongly significant: within-block mean $r = +0.0746$ against between-block -0.0282 , a gap of **+0.1027** against a null of -0.00003 ± 0.0077 , giving $z = 13.4$ with no permutation out of 5,000 reaching the observed value ($p < 0.0002$).

The blocks are not equally coherent. Ranking the diagonal gives environment +0.136, technology +0.066, education +0.054 and society/ethics +0.042: environment is more than three times as internally consistent as ethics. Among cross-block pairs, S–V is the one seam in the partition: at -0.004 it is the only pair not clearly separated, and it is also the strongest off-diagonal cell in the raw matrix (+0.256). T–V is the most separated pair (-0.046). This reproduces the ordering of the respondent-level block-score correlations computed independently on the raw file (T–E weakest at 0.29, S–V strongest at 0.70), though those are a different quantity—correlations between block average scores rather than means of item–item correlations—

and the two should not be compared numerically.

One reading caveat matters. The negative off-diagonal cells do not indicate opposition between topics. Row-centring forces every row to sum to zero, which pushes the grand mean of all 1,770 pairwise correlations slightly negative; measured, it is -0.0038 . The off-diagonal values should be read relative to each other and to the diagonal, not as evidence that technology views oppose environmental ones. The permutation null is unaffected, since observed and null statistics are computed on the same shifted matrix.

Implication. The survey’s thematic design is real and should be carried into the interpretation of every later partition: a detected community that aligns with a block is recovering something the instrument built in, and a community that cuts across blocks is the more informative object. The uneven diagonal sharpens this. Environmental attitudes travel as a package—agreement with one environmental item predicts agreement with others—whereas ethics items are close to independent of one another, so respondents hold ethical positions issue by issue rather than as a doctrine. The society–environment seam is where a genuine cross-block community is most likely to ap-

Threshold sweep on the MP-denoised item network

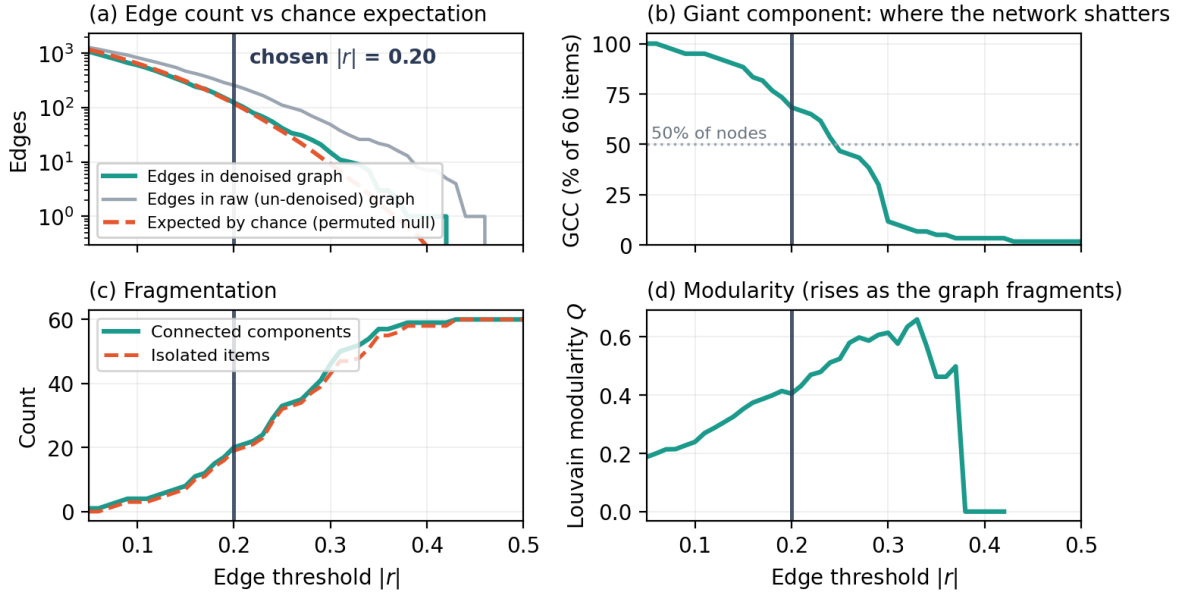


Figure 8: Graph 7. The signed item network rebuilt at all 46 cutoffs from 0.05 to 0.50. (a) Edge count against the chance expectation from Graph 4. (b) Giant component, which falls below two thirds of the items shortly after the chosen cutoff. (c) Components and isolates. (d) Louvain modularity, which rises to 0.63 only once the graph has broken into 51 pieces. The vertical line marks the selected $|r| > 0.20$.

pear, and the community structure reported in Graph 10 should be checked against that expectation rather than treated as a discovery if it reproduces it.

4.10 Graph 10: Communities against null models

[TO BE COMPLETED — Graph 10 — owner: P3]

Compare observed modularity and clustering against three nulls: a degree-preserving rewiring [6], an Erdős–Rényi graph, and a column-permutation null that regenerates the entire pipeline from scrambled responses.

Reference values (respondent network): modularity $0.408 - 0.423$ and clustering 0.202 observed; rewiring null 0.329 ± 0.011 ($z \approx 7$) and 0.091 ± 0.013 ($z \approx 8.9$), both passed; permutation null 0.403 ± 0.017 ($z \approx 1.15$), not passed.

Findings + implication: to be written. The permutation null is the decisive one: rewiring and ER take the finished graph as given and cannot detect transitivity introduced by the correlation-and-threshold construction itself. If modularity clears rewiring but not permutation, the community structure is largely a construction artefact and must be reported as such — this is a finding, not a failure.

4.11 Graph 11: Structural balance

[TO BE COMPLETED — Graph 11 — owner: P3]

Fraction of balanced triads in the signed item network as a function of threshold, against a sign-permutation null [2].

Reference values: balanced-triad fraction 83% at $|r| > 0.15$, 96% at 0.20, 100% at 0.25, against roughly 50% under the null.

Findings + implication: to be written. That the fraction strengthens as the threshold tightens is itself part of the evidence, since a spurious effect would not sharpen as weak edges are removed.

4.12 Graph 12: Opinion dynamics

This is the only analysis involving an actual dynamical process; everything above describes a fixed dataset. Because each opinion is one respondent's answer to one item, the vertices here are respondents, not items, so a second network is required.

The respondent network. From the row-centred matrix we compute the 91×91 Pearson correlation between respondents and connect each respondent to their $k = 4$ most similar peers, symmetrising by union. Union rather than mutual k NN is deliberate: mutual k NN

Graph 9: topic-block connectivity, 91 respondents $\times$ 60 items

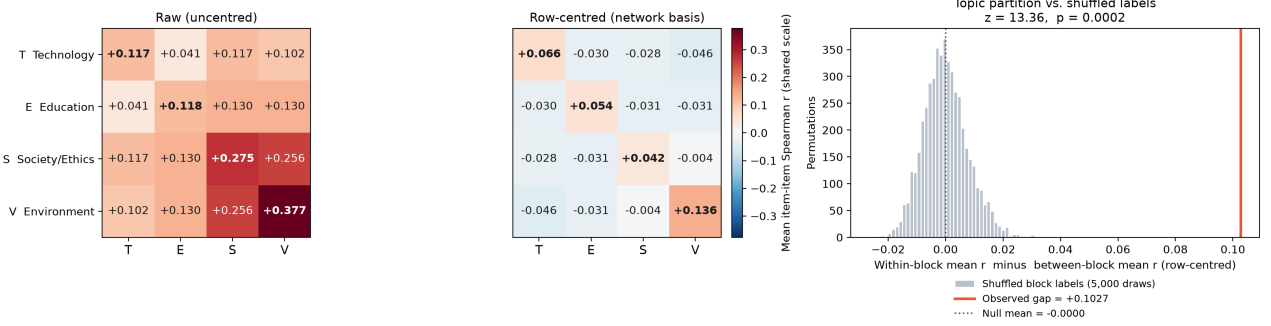


Figure 9: Graph 9. Topic-block connectivity, 91 respondents $\times$ 60 items. Left: mean item-item Spearman correlation within and between the four thematic blocks on the raw matrix—every cell positive, the design invisible under acquiescence. Centre: the same collapse on the row-centred matrix, sharing a colour scale; the diagonal turns positive and all off-diagonal cells negative. Diagonal cells average the 105 distinct within-block pairs, excluding self-correlations. Right: null distribution of the within-minus-between gap under 5,000 reshufflings of the T/E/S/V labels with the correlation matrix held fixed. The observed gap of +0.1027 ($z = 13.4$, $p < 0.0002$) is not approached by any permutation.

fragments this data badly, and disconnected components cannot merge at any confidence threshold, which would inflate the final cluster count for reasons unrelated to the dynamics. The result has 91 nodes, 298 edges, mean degree 6.55, and forms a single connected component.

The model. Opinions $x_i \in [0, 1]$ are seeded from item E03 (compulsory attendance, SD 1.12), rescaled by $s(x) = (x + 2)/4$. At each step a random edge (u, v) is drawn, and if the two opinions lie within the confidence threshold they move to their midpoint [4]:

$$|x_u - x_v| < \varepsilon \implies x_u, x_v \leftarrow \frac{1}{2}(x_u + x_v). \quad (2)$$

We sweep ε from 0.05 to 0.60 in steps of 0.05, with 20 seeded repetitions of up to 60,000 interactions each, terminating early on convergence, and count final clusters by rounding to two decimals.

Findings. Results appear in Figure 10 and Table 3. Cluster count is exactly 5.00 ± 0.00 for every $\varepsilon \leq 0.25$, then falls to 4.45 at $\varepsilon = 0.30$, 3.35 at 0.40, 2.00 at 0.50, approaching consensus near 0.55–0.60.

The plateau is an artefact, and provably so. The specification for this analysis requires stating that the low- ε plateau is partly an artefact of the discrete five-point grid. In fact the claim can be made exactly. Under $s(x) = (x + 2)/4$ the five categories map to $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, so any two distinct initial opinions differ by at least $\frac{1}{4}$. For any $\varepsilon \leq 0.25$ the condition $|x_u - x_v| < \varepsilon$ is therefore never satisfied by a pair with distinct opinions: the initial state is a fixed point and

the final count equals the number of occupied categories, on any graph whatsoever.

Three observations confirm this empirically. The plateau has zero variance across 20 randomly seeded runs, which no genuine emergent outcome would. A complete graph—a maximally different topology—returns the identical 5.00 ± 0.00 over the same range. And re-running with seeds jittered by ± 0.125 , which destroys the grid while preserving the distribution’s shape, removes the plateau entirely, yielding $35.95 \rightarrow 26.35 \rightarrow 21.05 \rightarrow 14.20 \rightarrow 8.95$ clusters across $\varepsilon = 0.05 \dots 0.25$. The five clusters below $\varepsilon = 0.25$ are the original response categories failing to merge, not opinion groups forming, and the break in the curve at $\varepsilon = 0.30$ is arithmetic rather than a discovered threshold.

What is not an artefact. The substantive result lies above $\varepsilon = 0.25$. Comparing the real network against an identically seeded complete graph—the same dynamics with topology removed—the well-mixed population collapses to about 2 clusters across $\varepsilon = 0.30$ –0.45, while the sparse k NN network sustains 3.15–4.45. Network structure therefore sustains opinion diversity that a well-mixed population loses at the same tolerance: when respondents interact only with their nearest neighbours in opinion space, locally cohesive groups stabilise before global consensus can propagate. This is the one result in the figure that is neither a scale artefact nor a restatement of the input distribution, and it is what justifies studying the process on a network at all.

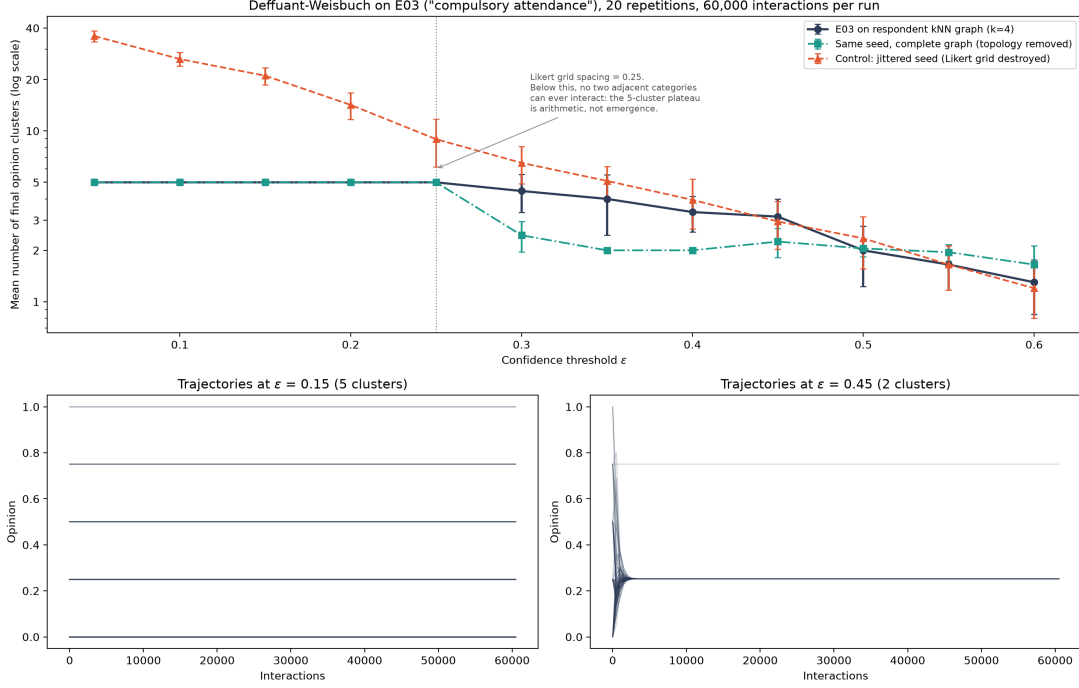


Figure 10: Graph 12. Top: mean final cluster count against confidence threshold ϵ on a log scale, with error bars over 20 repetitions. The k NN network and the complete graph coincide exactly below $\epsilon = 0.25$ —the signature of an arithmetic artefact—and diverge above it, where topology matters. The jittered control destroys the response grid and removes the plateau entirely. Bottom: individual opinion trajectories at a low and a high ϵ , showing frozen categories against rapid collapse.

Table 3: Deffuant–Weisbuch final cluster counts (mean $\pm$ sd over 20 repetitions). The plateau at $\epsilon \leq 0.25$ is forced by the response grid and appears identically on any topology; the gap between the k NN and complete graphs above it is the genuine topological effect.

ϵ	k NN graph	Complete graph	Jittered control
≤ 0.25	5.00 ± 0.00	5.00 ± 0.00	$36.0 \rightarrow 9.0$
0.30	4.45 ± 1.12	2.45 ± 0.50	6.50 ± 1.60
0.35	4.00 ± 1.55	2.00 ± 0.00	5.10 ± 1.09
0.40	3.35 ± 0.79	2.00 ± 0.00	3.95 ± 1.28
0.45	3.15 ± 0.85	2.25 ± 0.43	2.95 ± 0.92
0.50	2.00 ± 0.77	2.05 ± 0.22	2.35 ± 0.79
0.55	1.65 ± 0.48	1.95 ± 0.22	1.65 ± 0.48
0.60	1.30 ± 0.46	1.65 ± 0.48	1.20 ± 0.40

5 Results and Discussion

5.1 What the data supports

Superficial consensus with genuine local structure. The headline finding is not polarisation. Agreement runs at 79.8% across every block and the leading principal component explains only about 8% of variance, so there is no evidence of two opposed camps. What ex-

ists instead is a broad agreeable consensus with real but local disagreement concentrated in specific, identifiable places.

Disagreement is domain-specific, not dispositional. Environment and society are effectively settled; technology and education are not. Within education the cohort is precise about what it rejects—compulsory attendance and written examinations—while endorsing curriculum reform and problem-solving at above 94%. Within technology the objection tracks delegated consequential judgment (medical diagnosis, autonomous vehicles) rather than automation as such.

Most apparent structure was response style. Row-centring removes 87% of the item pairs exceeding $|r| \geq 0.30$. Any analysis of this questionnaire that skips this step will find dense, uniformly positive structure and will be measuring agreeableness.

Topology has a measurable dynamical consequence. In the bounded-confidence model, the sparse respondent network sustains noticeably more opinion clusters than a well-mixed population at the same tolerance,

over the range where the dynamics are genuinely active.

5.2 What the data does not support

Four fifths of the correlation matrix is noise. Section 4.5 puts a number on how much of this matrix is interpretable: exactly three eigenvalue modes exceed the Marchenko–Pastur band, carrying 19.3% of total variance. The remaining 80.7% is what 86 respondents answering 60 items generate by chance. Any claim resting on a single correlation, or on the fine structure of the full matrix, is unsupported.

The community structure does not survive its own null. This is the sharpest negative result in the report, and which null model is used decides it. A correlation-derived network has transitivity built into its construction: if A correlates with B and B with C, A and C are pulled toward correlation automatically, and that alone manufactures clustering and modularity in data with no association whatsoever. Degree-preserving rewiring [6] and Erdős–Rényi nulls both take the finished graph as given, so neither can detect structure that the construction itself created. Only a permutation null—destroying association in the raw responses, then regenerating the entire chain of centre, correlate, threshold and measure—tests the claim actually being made. On the respondent network (86 nodes, 286 edges, one component, symmetrised k NN at $k = 4$), observed modularity is 0.416 and average clustering 0.202:

Null model	Mean	SD	z
Degree-preserving rewiring	0.329	0.011	7.74
Erdős–Rényi $G(n, m)$	0.332	0.011	7.79
Column permutation	0.400	0.019	0.79

Against the two graph-level nulls the structure looks overwhelming, at z approaching 8. Against the null that reproduces the construction it vanishes ($p = 0.20$). We report this as a finding rather than a failure: those communities are an artefact of correlating and then applying a neighbourhood rule. It is also the reason the item network, thresholded directly on a measured quantity, is the primary object of this report and the respondent network is used only for the dynamics of Section 4.12.

And therefore, not polarisation. Three cross-cutting modes carrying under a fifth of variance, communities that fail their own null, and 79.8% agreement across every block describe a cohort in broad consensus with local, domain-specific disagreement. They do not describe two camps.

[TO BE COMPLETED — Results and Discussion — integration, owner: P3]

Items (i) and (ii) are now answered above from Graphs 5 and 10. Still outstanding:

- (iii) whether the detected communities align with the four thematic blocks or cut across them (Graphs 6 and 9);
- (iv) whether structural balance strengthens with threshold (Graph 11).

5.3 Methods considered and rejected

Documenting what did not work is part of the analysis. Four alternatives were evaluated and set aside.

Recoding NC as Neutral imposes a systematic shift of about 0.92 scale points at those cells and asserts a response that was explicitly withheld. Dropping NC users removes roughly a fifth of the sample to eliminate 37 cells and is the only NC treatment that substantially disturbs the recovered partition (AMI 0.337). Excluding respondents by missing-item count was rejected because, once the five empty records are removed, no substantial population of badly-incomplete respondents remains: the rest are 95.6% complete, and model-based recovery is preferable to discarding usable respondents. Mutual k NN for the respondent network fragments the graph into many components, which would inflate cluster counts in the dynamics for reasons unrelated to the process being studied; union k NN yields a single connected component.

Four further alternatives were measured rather than argued about.

Percentile thresholding at p90 fixes the edge count by construction, so it cannot say whether any edge beats chance — the objection is structural, not empirical. Measured, it behaves very differently on the two networks: on the item network it implies $|r| > 0.229$ and leaves 177 edges in a single component, while on the respondent network it implies $|r| > 0.504$ and leaves 366 edges across 22 components with 20 isolates.

Raw uncentred similarity leaves 696 of 1,770 pairs above $|r| > 0.20$ — a 39%-dense graph in which 83% of all correlations are positive. This is the same acquiescence effect quantified in Section 4.3, restated at the threshold actually used.

Column-median instead of column-mean imputation was tested rather than assumed: across the 47 missing cells it moves the correlation SD from 0.1363 to 0.1361 and the edge count at $|r| > 0.20$ from 258 to 260, and leaves the number of significant eigenvalue modes unchanged at three. The choice is immaterial, which is itself the argument against anything heavier.

Retaining all 91 non-empty respondents rather than dropping the ten with more than twelve missing items is a genuine sensitivity, and we report it because it changes a headline. At $T = 91$, $Q = 1.517$ and $\lambda_+ = 3.283$, and four modes clear the band rather than three,

carrying 25.6% of variance. The extra mode is supported disproportionately by the 242 cells recovered by the imputation model of Section 3.2, whose held-out accuracy is 47.6%. Graphs 4, 5 and 7 therefore use the 86-respondent matrix, and the “exactly three dimensions” result should be read as conditional on that filter rather than as a property of the cohort.

5.4 Limitations

Four limits bound every claim above. The response ceiling means items with means above roughly 4.5 measure social desirability more than preference strength, and because few items are reverse-worded, acquiescence and a genuine ceiling cannot be fully separated. The sample is $n = 96$ from a single institution, so differences of a few percentage points between statements are noise and only the large gaps are safely above sampling variation. Dropout is monotone, so later blocks rest on a slightly smaller base than earlier ones. And the file carries no demographic fields, so no finding can be attributed to any subgroup without additional joined data.

A methodological limitation deserves separate mention: correlation stability under perturbation does not imply partition stability. Table 1 shows correlation matrices agreeing at $r > 0.94$ across NC treatments whose community assignments agree at only AMI 0.337. Conclusions drawn at the level of communities require their own stability analysis, by respondent-level bootstrap resampling, before being treated as findings.

6 Conclusion

We presented an end-to-end pipeline from a raw ordinal questionnaire to an opinion network and a dynamical process on it. Sanitisation preserves the distinction between abandonment and deliberate non-response, retains all 91 non-empty respondents, and recovers 242 missing cells by question-wise regularised proportional-odds ordinal regression, which beats a modal baseline under both scattered and structural artificial masking. Row-centring then removes the acquiescence component that would otherwise dominate every correlation, an intervention whose magnitude we quantify rather than assume: it eliminates 87% of item pairs correlated at $|r| \geq 0.30$. On the resulting matrix we build signed item and respondent networks and run bounded-confidence dynamics, showing by direct argument and controlled experiment that the small- ε cluster plateau is forced by the five-point response grid, while the diversity sustained by the sparse network relative to a well-mixed population is a genuine topological effect.

On that matrix we then measured how much is interpretable at all. A permutation null that regen-

erates the whole pipeline from independently shuffled columns puts the noise floor at 0.108 against an observed correlation spread of 0.136, and Marchenko–Pastur filtering leaves exactly three modes above the band, carrying 19.3% of variance. The edge threshold was then selected from a 46-point sweep by a stated rule rather than convention, at $|r| > 0.20$, where the giant component still spans 68% of the items.

The same null framework delivers the report’s sharpest negative result. The respondent network’s community structure clears a degree-preserving rewiring null at $z = 7.74$ and an Erdős–Rényi null at $z = 7.79$, but fails the permutation null at $z = 0.79$ ($p = 0.20$): it is an artefact of correlating and then applying a neighbourhood rule, not a property of the responses. The characterisation that survives all of this is a cohort in superficial consensus with genuine but local structure — three modest cross-cutting dimensions, not two camps.

[TO BE COMPLETED — Conclusion — final integration]
Still to add once Graphs 6 and 11 land: the number of item-network communities and whether they align with the four thematic blocks, and whether structural balance strengthens with threshold.

Individual Contribution

Table 4: Contribution by team member.

Member	Contribution
Arijeet Paul	Data sanitation and missingness analysis; ordinal-regression imputation and its masking validation; row-centring; Graphs 1, 2, 3 and 12; respondent k NN network; repository structure and run_all.py.
Shreyash Lohare	Noise-floor analysis and the permutation null framework (src/common/nulls.py); Marchenko–Pastur spectral denoising; threshold sweep and selection; the three-null comparison of Section 5.2; sensitivity analysis of the rejected methods in Section 5.3; shared configuration and the spec-conformant analysis matrix (src/common/); Graphs 4, 5 and 7; this section and Section 4.4–4.7. Commits tagged [P2].
Dev Patel	[TO BE COMPLETED — Graphs 6, 10, 11]

[TO BE COMPLETED — Individual contribution]
Each member should replace their row with specific, auditable contributions (modules written, figures produced, sections drafted). git log provides the record.

References

- [1] Vincent D Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008.
- [2] Dorwin Cartwright and Frank Harary. Structural balance: a generalization of heider’s theory. *Psychological Review*, 63(5):277, 1956.
- [3] Lee J Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, 1951.
- [4] Guillaume Deffuant, David Neau, Frederic Amblard, and Gérard Weisbuch. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3(01n04):87–98, 2000.
- [5] Vladimir A Marčenko and Leonid A Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457, 1967.
- [6] Sergei Maslov and Kim Sneppen. Specificity and stability in topology of protein networks. *Science*, 296(5569):910–913, 2002.
- [7] Peter McCullagh. Regression models for ordinal data. *Journal of the Royal Statistical Society: Series B*, 42(2):109–127, 1980.
- [8] Mark EJ Newman. Modularity and community structure in networks. *Proceedings of the National Academy of Sciences*, 103(23):8577–8582, 2006.
- [9] Delroy L Paulhus. Measurement and control of response bias. *Measures of Personality and Social Psychological Attitudes*, pages 17–59, 1991.
- [10] Donald B Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976.